

AI Cyber Defense Ops

Detection Engineering MCP Server

Capstone Project Report

A practical application of Claude ecosystem tools for detection engineering

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Course: AI Cyber Defense Ops (Module 10+: Purple Team & Multi-Agent Workflows)

Date: July 30, 2026

Status: Ready for Academic Review

Executive Summary

This capstone project implements a **Model Context Protocol (MCP) server** that provides Claude with integrated detection engineering capabilities for defensive security operations. The system combines log analysis, threat intelligence processing, and detection rule management into a unified platform for security investigations.

By leveraging Claude's MCP framework, Slash Commands, Skills, and multi-agent workflows, this project demonstrates how the Claude ecosystem can be applied to operationalize detection engineering — from ingesting threat intelligence to validating detection rules through purple team exercises.

Key Accomplishments

-  **MCP Server Architecture:** Built a FastMCP-based server exposing tools and resources for security analysis
-  **Detection Knowledge Base:** Integrated Sigma detection rules with ATT&CK technique mappings
-  **Threat Intelligence Pipeline:** Automated extraction of TTPs, IOCs, and simulation plans from threat reports
-  **Multi-Source Investigation:** Correlated endpoint (Sysmon/Windows Security) logs with structured analysis
-  **Coverage Analysis:** Quantified detection gaps and validated rule coverage against real attack scenarios
-  **Purple Team Integration:** Executed and analyzed atomic red team tests (T1082, T1069.002, T1136.001)

Project Goals & Accomplishments

Original Goals

This capstone project aimed to build a **detection engineering platform** that would:

1. Expose Sigma rules as browsable resources to Claude
2. Map detection rules to MITRE ATT&CK techniques
3. Query detection coverage for specific techniques or tactics
4. Combine endpoint telemetry with threat intelligence
5. Process threat intel reports into actionable simulation plans
6. Query Splunk to correlate Windows Security and Sysmon logs into unified attack narratives

Completed Objectives

Core Platform Delivery

Objective	Status	Evidence
MCP server deployment	 Complete	server.py with FastMCP framework, exposing 6+ tools and 5+ resource types
Sigma rule library	 Complete	20+ production-ready rules covering Credential Access and Reconnaissance tactics
ATT&CK mapping system	 Complete	mappings/attack_techniques.py with 50+ technique references
Tool suite	 Complete	Coverage analysis, rule suggestion, Sysmon parsing, Splunk query execution
Investigation workflows	 Complete	/query, /hunt, /ingest-ti commands with Obsidian-ready output

Coverage Expansion: FSB Center 16 Campaign

Gap Identified: Initial rule set covered only Windows endpoint Credential Access. No network-appliance, SNMP, or non-Windows telemetry support.

Response: /ingest-ti against CISA AA26-194A identified 15 ATT&CK techniques used by FSB Center 16. Gap remediation added:

Rule	Techniques	Severity	Status
Outbound TFTP Transfer to Unrecognized Destination	T1048.003, T1041	High	Experimental
Inbound Traffic to Cisco Smart Install Port	T1190	High	Experimental
SNMP Set-Request Targeting Cisco Config-Copy	T1602.001, T1595.001	Medium	Experimental

Outcome: 3 new Sigma rules + 5 new ATT&CK mappings added. Structural foundation established for future network-device-targeting advisories.

Purple Team Loop Execution (July 26–27, 2026)

Complete End-to-End Validation on Your Machine

Step 1: Threat Intelligence Ingest

Campaign: Huntress Report — ClickFix + Matanbuchus 3.0 + AstarionRAT

Extracted 10+ techniques across Discovery, Execution, Persistence, and Command & Control tactics.

Step 2: Atomic Test Mapping

Execution Scope (This Cycle): Discovery-phase techniques only

Technique	Atomic Test	Expected Detection
T1082 — System Information Discovery	Invoke-AtomicTest T1082	Sysmon EventID 1: systeminfo.exe + reg.exe
T1069.002 — Domain Groups Discovery	Invoke-AtomicTest T1069.002	Sysmon EventID 1: net.exe (4 variants)

Step 3: Execution on Your Development Machine (HOST-A)

Date: July 26, 2026 20:51:56–20:52:03 PDT

```
# T1082 - System Information Discovery
systeminfo & reg query HKLM\SYSTEM\CurrentControlSet\Services\Disk\Enum

# T1069.002 - Domain Groups Discovery (4 variants)
net localgroup
net group /domain
net group "enterprise admins" /domain
net group "domain admins" /domain
```

Result: All 5 commands executed and logged. Domain queries returned expected errors (not domain-joined) but still generated process-creation telemetry.

Step 4: Hayabusa Scanning

File	Detections	Notes
Security.evtx (21MB)	9,813 (9,660 low, 153 info)	No medium+ Sigma hits, audit noise only
Sysmon.evtx (66MB)	37,156 @ critical	Large file limitation: lower severity scans timeout

Step 5: SIEM Validation (Splunk)

Technique	Result	Status	Finding
T1082 (System Info)	Confirmed in Sysmon	 Detected	systeminfo.exe + reg.exe at exact timestamp
T1069.002 (Domain Groups)	Confirmed in Sysmon (4/4 commands)	 Detected	All net.exe variants captured
Windows Security 4688	NOT PRESENT	 Gap	"Audit Process Creation" not enabled

Step 6: Gap Analysis

Gap	Type	Remediation
Windows Security 4688 disabled	Coverage Gap	Enable "Audit Process Creation" via auditpol
Sysmon ATT&CK tagging mismatch	Operational	Reconcile local Sysmon config before using for dashboards
Hayabusa large-file timeout	Tooling	Chunk scans by time range or raise buffer limits

Step 7: Reporting & Documentation

purple-team-exercise-report-2026-07-26.docx (Analyst Report — 3 Pages)

Location: components/purple-team/reports/2026-07-26/purple-team-exercise-report-2026-07-26.docx

Audience: Security analysts, detection engineers, SOC teams

Contents:

Page	Section	Content
1	Executive Overview	Campaign summary (ClickFix/Matanbuchus), test scope (T1082, T1069.002 Discovery phase), objectives and methodology
1-2	Execution Timeline	Detailed chronology of purple team execution on HOST-A (July 26, 20:51:56–20:52:03 PDT): command sequence, timestamps, expected vs. actual telemetry, process IDs
2	SIEM Validation Results	Splunk query results for Sysmon and Windows Security events: T1082 detection (systeminfo.exe + reg.exe), T1069.002 detection (net.exe variants), EventCode 4688 audit gap documented
2-3	Hayabusa Findings	EVTX scan results: Security.evtx (21MB, 9,813 detections — audit noise), Sysmon.evtx (66MB, 37,156 detections @ critical), known MCP buffer limitation documented
3	Gap Analysis & Remediation	Coverage assessment (Telemetry 100%, Windows Security 0% due to audit gap, Sysmon 100%), ATT&CK tagging mismatch between Sysmon config (T1033/T1018) and exercise mapping (T1082/T1069.002), recommended next steps
3	Appendix	SPL queries used, EVTX file manifest, tool version info, contact information for follow-up

Location: components/purple-team/reports/2026-07-26/purple-team-exercise-leadership-2026-07-26.pptx

Audience: CISO, senior management, executive stakeholders

Contents:

Slide	Title	Content
1	Title Slide	Purple Team Exercise 2026-07-26 ClickFix + Matanbuchus Campaign Simulation Date, stakeholders
2	Campaign Overview	Threat description (ClickFix fake driver, Matanbuchus loader, AstarionRAT), attack chain diagram, real-world relevance, why simulation matters
3	Exercise Scope	Discovery phase focus (T1082, T1069.002), test environment (HOST-A physical workstation, sanitized), execution timeline, constraints and out-of-scope items
4	Detection Coverage Metrics	Overall Coverage: 8% of ClickFix/Matanbuchus techniques Discovery Tactic: 50% Telemetry Capture: 100% (Sysmon confirmed) Windows Security: 0% (audit gap) Breakdown by tactic with color-coded status badges
5	KEY FINDING 1: Real Coverage Gap	[RED BADGE] Windows Security EventCode 4688 (Process Creation) is NOT enabled on monitored endpoints Impact: Discovery-phase commands detected in Sysmon but invisible to Windows Security auditing Remediation: Enable "Audit Process Creation" via Group Policy / auditpol
6	KEY FINDING 2: Tooling Caveat	[AMBER BADGE] Hayabusa scanner encounters MCP buffer timeout on large EVTX files (66MB Sysmon.evtx) at lower severity thresholds Impact: Cannot validate full rule set against large datasets Workaround: Chunk scans by time

Slide	Title	Content
		range Note: Does NOT affect telemetry capture or SIEM validation
7	Recommendations & Next Steps	Priority 1: Enable Windows Security Process Creation auditing (immediately addressable, high impact) Priority 2: Reconcile Sysmon ATT&CK tagging (local config uses T1033/T1018, exercise uses T1082/T1069.002) Priority 3: Extend testing to multi-host environment (requires domain setup) Future: Lateral movement techniques (T1021.x, T1570)
8	Conclusion	Telemetry capture validated at 100% for tested techniques Real coverage gap identified and remediation path clear Tooling limitations documented and manageable Ready to expand to broader attack chains once prerequisites met

Design Note: Per instructions, slides 5 and 6 keep the real coverage gap (Finding 1, red badge) visually and textually separate from the tooling caveat (Finding 2, amber badge) to avoid conflating operational gaps with tool limitations.

Step 8: Vectr Tracking

Formatted for security tracking system: Campaign, Test Cases, detection outcomes per data source.

T1136.001 Validation (July 28–29, 2026)

Local Account Creation Test (Disposable VM — HOST-B)

Test: Create local account, promote to Administrators, cleanup

Action	Command	Detection
Create account	net user /add "T1136.001_Admin" "T1136_pass"	 Windows Security 4720
Escalate privileges	net localgroup administrators ... /add	 Windows Security 4732
Verify creation	net user T1136.001_Admin	Sysmon EventID 1
Delete account	net user /del "T1136.001_Admin"	Sysmon EventID 1
Verify deletion	net user T1136.001_Admin (verify gone)	Sysmon EventID 1

Conclusion: T1136.001 is fully detectable. Account created, tested, deleted — zero lateral movement detected. Perfect validation of Windows Security event auditing.

Key Metrics & Summary

Metric	Value	Status
Sigma Rules Written	20+	✓ Production
ATT&CK Techniques Mapped	50+	✓ Production
Gap Remediation Rules	3	● Experimental
Threat Intel Reports Ingested	1 (CISA AA26-194A)	✓ Tested
Credential Access Hunts	7	✓ Covered
Hunts with Data Gaps	2	⚠ Inconclusive
Purple Team Tests Executed	3 techniques	✓ Validated

Deliverables

Code & Configuration

- `server.py` — MCP server (FastMCP, 500+ lines, 6 tools, 5 resources)
- `rules/` — 20+ Sigma YAML rules
- `mappings/` — ATT&CK technique database (50+ techniques)
- `.claude/commands/` — `/query`, `/hunt`, `/ingest-ti`
- `.claude/agents/` — Subagents (endpoint, cloud, atomic mapper)

Documentation

- `CLAUDE.md` — Project specification and workflow guide

- `investigations/` — 9 Obsidian-ready hunt notes
- `analysis/` — Threat intel analyses and gap-remediation writeups

Reports & Analysis

- `sysmon-report-2026-07-29-sanitized.md` — 24-hour telemetry (sanitized)
- `gap-remediation-fsb-center16-network-appliance-coverage.md` — Gap analysis
- `ti-2026-07-20-fsb-center16-router-targeting.md` — Threat intel output
- `purple-team-exercise-report-2026-07-26.docx` — Analyst documentation
- `purple-team-exercise-leadership-2026-07-26.pptx` — CISO-ready deck

Conclusion

This capstone project delivers a **production-ready detection engineering platform** combining unified log analysis, threat intelligence automation, detection knowledge management, and purple team integration. The platform is extensible and ready for new telemetry sources (network flows, cloud logs) as they become available.

Outcome: A detection engineering capability that turns threat intelligence into testable, measurable security controls.

Sanitization Notice: All personal identifiers and credentials have been sanitized.

Machine names: HOST-A (physical), HOST-B (VM) | Usernames: USER-A, USER-B

For questions, refer to CLAUDE.md project specification.

